Emergent Born Rule via Kinematically Constrained Quantum Scrambling

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Abstract

The Born rule is universally postulated as an irreducible axiom of standard quantum mechanics, lacking a dynamical or first-principles derivation. In this paper, we provide a rigorous derivation of the Born rule by embedding local quantum measurements within a kinematically constrained, fully scrambled many-body environment. By projecting late-time dynamics onto a Haar-random subspace, we show via Page's theorem that local subsystems are driven exponentially close to a maximally mixed state. This maximal entanglement naturally activates Zurek's environment-assisted invariance (envariance). Relativistic causality - specifically the no-signaling theorem - rigidly enforces equal probabilities for these fine-grained microstates without invoking prior probabilistic assumptions. Coarse-graining then yields the Born rule. By addressing the principal objections to envariance-based derivations, including the circularity objection and the preferred-basis problem, this framework successfully shifts the origin of quantum probability from an ad hoc axiom to an emergent consequence of causality and the classical geometry of scrambled states.

1 Introduction

Two of the most profound conceptual tensions in theoretical physics concern the foundational origin of quantum probability and the role of ensemble randomness in quantum chaos. In standard quantum mechanics, the Born rule - the prescription that a measurement outcome occurs with probability $p_k = |\psi_k|^2$ - is introduced as an irreducible axiom with no dynamical derivation [1]. Meanwhile, explaining the universal chaotic signatures of the Spectral Form Factor (SFF) in many-body systems has traditionally relied on drawing Hamiltonians from random matrix ensembles [2, 3]. This statistical averaging procedure introduces fundamental

randomness that appears to conflict with the strict unitary evolution mandated for isolated quantum systems.

In this article, we demonstrate that these two apparently unrelated problems admit a common resolution within a single kinematic framework. The central postulate is that the fundamental Hamiltonian of a complex system is strictly fixed and non-random. The appearance of ensemble randomness in the SFF emerges as a kinematic consequence of projecting unitary evolution onto a Haar-random subspace of late-time, fully scrambled states - a subspace whose geometry is set by the specific boundary conditions of the universe instance we inhabit.

More precisely, our program proceeds in three stages. In Sec. 2, we define the kinematic constraint, construct the effective projected Hamiltonian, and derive the spectral statistics of the compressed ensemble. Except for the initial setup, rest of the section intends to reproduce the well-known and well-established Wigner-Dyson results, from which the dip-ramp-plateau SFF follows by Fourier transform of the sine-kernel two-point function. In Sec. 3, we partition the total system into a local probe and a many-body environment, and invoke Page's theorem to show that the kinematic constraint drives the local density matrix to a maximally mixed state. In Secs. 4 and 5, we give a rigorous, step-by-step derivation of the Born rule using Zurek's envariance, with no-signaling playing the role of the key physical constraint. Sec. 6 systematically addresses the major objections to this framework.

Our framework builds on the substantial literature concerning quantum chaos, thermalization, and the measurement problem. The SFF has been established as a central diagnostic of quantum chaos in black holes [2], holographic systems [4], and many-body systems [5]. The connection between scrambling and decohering environments has been explored in the context of Hayden-Preskill recovery [6] and black hole information [7]. Envariance-based derivations of the Born rule have been developed by Zurek [14, 15] and subsequently analyzed and debated extensively [16, 17, 18]. Our contribution is to provide the physical structure justification - kinematically constrained scrambling - that simultaneously justifies both the random matrix ensemble and the envariance argument.

2 Kinematically Constrained Scrambling and the Spectral Form Factor

2.1 Setup and Kinematic Ansatz

Let $\mathcal{H}$ be the full Hilbert space of an isolated many-body system, with $\dim \mathcal{H} = L$. We work in the thermodynamic regime $L \to \infty$. We postulate two fixed, non-random operators: a pre-scrambling Hamiltonian H_{base} governing early-time deterministic dynamics, and a boundary Hamiltonian H_{post} with a strictly discrete spectrum $\{E_n\}_{n=1}^{L}$ governing late-time equilibrium degrees of freedom.

Definition 2.1 (Kinematic Constraint)**.** The late-time scrambled sector is an N-dimensional subspace $\mathcal{V}_\alpha \subset \mathcal{H}$, with $N \sim e^{S_{\text{th}}}$, defined by the instance-dependent projector

$$\Pi_\alpha = V_\alpha P_0 V_\alpha^\dagger, \tag{1}$$

where P_0 is a fixed canonical reference projector of rank N, and $V_\alpha \in G$ is a Haar-random *unitary*. The first N columns of V_α form the $L \times N$ Haar-random *isometry* $\Phi_\alpha : \mathbb{C}^N \to \mathbb{C}^L$, satisfying $\Phi_\alpha^\dagger \Phi_\alpha = I_N$ and $\Phi_\alpha \Phi_\alpha^\dagger = \Pi_\alpha$. The symmetry group $G \in \{U(L), O(L), Sp(L)\}$ is selected by the time-reversal symmetry class of H_{post}, corresponding to GUE, GOE, and GSE universality classes respectively.

The total Hamiltonian governing instance α is strictly block-diagonal:

$$H(\alpha) = (I - \Pi_\alpha)\, H_{\text{base}}\, (I - \Pi_\alpha) + \Pi_\alpha\, H_{\text{post}}\, \Pi_\alpha. \tag{2}$$

At late times $t \gg t_{\text{Th}}$ (the Thouless time), the first term is exponentially suppressed, and dynamics are controlled entirely by the effective projected Hamiltonian

$$h_\alpha \equiv \Pi_\alpha H_{\text{post}} \Pi_\alpha = V_\alpha\, \tilde{H}\, V_\alpha^\dagger, \tag{3}$$

where $\tilde{H} = P_0 H_{\text{post}} P_0$ denotes the $N \times N$ matrix obtained by restricting H_{post} to the reference subspace.

Remark 2.2. The effective compressed Hamiltonian governing the late-time sector is

$$\mathcal{H}_\alpha^{(N)} \equiv \Phi_\alpha^\dagger H_{\text{post}} \Phi_\alpha, \tag{4}$$

an $N \times N$ self-adjoint matrix whose eigenvalues $\{\epsilon_j(\alpha)\}_{j=1}^N$ depend non-trivially on the instance α through Φ_α. Crucially, higher powers of $\mathcal{H}_\alpha^{(N)}$ satisfy $(\mathcal{H}_\alpha^{(N)})^k = \Phi_\alpha^\dagger H_{\text{post}} (P_\alpha H_{\text{post}})^{k-1} \Phi_\alpha$, where $P_\alpha = \Phi_\alpha \Phi_\alpha^\dagger$ is the rank-N projector. Because $P_\alpha \neq I_L$, the matrix exponential does *not* factorize as $e^{-i\mathcal{H}_\alpha^{(N)} t} = \Phi_\alpha^\dagger e^{-iH_{\text{post}} t} \Phi_\alpha$. The spectral statistics of $\mathcal{H}_\alpha^{(N)}$ must therefore be derived via the resolvent, as detailed in Sec. 2.2.

2.2 Resolvent, Free Compression, and Spectral Statistics

Since the matrix exponential does not factor through the compression (Remark 2.2), the spectral statistics of $\mathcal{H}_\alpha^{(N)}$ must be derived at the level of the resolvent. Define the normalized Cauchy (Stieltjes) transform

$$G_\alpha(z) \equiv \frac{1}{N} \text{Tr}_N \left[\left(z\, I_N - \mathcal{H}_\alpha^{(N)} \right)^{-1} \right], \quad z \in \mathbb{C} \setminus \mathbb{R}. \tag{5}$$

The Neumann series expansion of the resolvent reads

$$\left(z - \mathcal{H}_\alpha^{(N)} \right)^{-1} = \frac{1}{z} \sum_{k=0}^{\infty} \frac{1}{z^k} \left(\Phi_\alpha^\dagger H_{\text{post}} \Phi_\alpha \right)^k, \tag{6}$$

where the k-th power is $\Phi_\alpha^\dagger H_{\text{post}} (P_\alpha H_{\text{post}})^{k-1} \Phi_\alpha$, with $P_\alpha = \Phi_\alpha \Phi_\alpha^\dagger$ appearing explicitly between each factor of H_{post}. In contrast to the exponential, the resolvent correctly retains these projectors. Taking the Haar average $\overline{G_\alpha(z)}$ requires evaluating integrals of products of P_α matrix elements, handled by the Weingarten formula of [8] applied to $V_\alpha \in G$. In the large-L limit with $N/L \to c$, the Weingarten sum is dominated by *non-crossing* pair partitions; crossing contributions are suppressed by $O(L^{-1})$ [36]. Resumming the non-crossing contributions over all orders k yields:

Theorem 2.3 (Free Compression [8, 36])**.** *Let H_{post} be a deterministic self-adjoint $L \times L$ matrix whose empirical spectral measure $\mu_H = L^{-1} \sum_n \delta_{E_n}$ converges weakly to a compactly supported probability measure μ as $L \to \infty$. Let Φ_α be a Haar-random isometry with $N/L \to c \in (0, 1)$. Then the empirical spectral measure of $\mathcal{H}_\alpha^{(N)}$ converges almost surely to the deterministic*

measure μ_c whose Cauchy transform $G_c(z)$ satisfies the subordination equation

$$G_c(z) = \frac{1}{c}\, G_\mu\!\left(z - \frac{1-c}{c}\,\frac{1}{G_c(z)}\right), \tag{7}$$

where $G_\mu(w) = \int (w-\lambda)^{-1} d\mu(\lambda)$ is the Cauchy transform of μ.

The mean spectral density of the compressed sector is recovered by Stieltjes inversion: $\rho_c(\epsilon) = -\frac{1}{\pi c}\lim_{\eta\to 0^+} \operatorname{Im} G_c(\epsilon + i\eta)$. Equation (7) is the exact large-L resummation of the Weingarten series in (6); it correctly incorporates the $P_\alpha \neq I_L$ corrections that invalidate the time-domain approach.

2.3 Wigner-Dyson Statistics and the Dip-Ramp-Plateau

Theorem 2.3 establishes the mean spectral density ρ_c. The SFF depends on the *fluctuations* of eigenvalues $\{\epsilon_j(\alpha)\}$ around their mean positions, which are governed by a separate universality result.

Theorem 2.4 (Wigner-Dyson Universality [9, 37])**.** *For any fixed W in the N-dimensional symmetry group, $\mathcal{H}_\alpha^{(N)} \overset{d}{=} W^\dagger \mathcal{H}_\alpha^{(N)} W$. By the universality theorem for rotationally invariant matrix ensembles, the local bulk eigenvalue statistics of $\mathcal{H}_\alpha^{(N)}$ converge to the GUE (resp. GOE, GSE) sine-kernel statistics for $G = U(L)$ (resp. $O(L)$, $Sp(L)$), independently of the particular spectral measure μ of H_{post}.*

Given the sine-kernel two-point correlation in unfolded units $s = N\rho_c(\epsilon_1 - \epsilon_2)$,

$$R_{2,c}(s) = \rho_c^2\left[1 - \left(\frac{\sin \pi s}{\pi s}\right)^2\right], \tag{8}$$

the connected SFF follows by Fourier transform of the cluster function $T_2(s) = \rho_c^2 - R_{2,c}(s)$ [3]:

$$K_{\text{conn}}(t) = \begin{cases} |t|/(2\pi) & |t| \le t_H, \\ t_H/(2\pi) = N\rho_c & |t| > t_H. \end{cases} \tag{9}$$

The **linear ramp** for $t_{\text{erg}} \ll t \ll t_H$ is a direct consequence of eigenvalue repulsion in the Wigner-Dyson ensemble, rigorously established via Theorem 2.4. The **dip** at early times arises from the disconnected SFF $K_{\text{disc}}(t) = |\overline{Z(t)}|^2$, which decays by dephasing over the spectral bandwidth of ρ_c. The **plateau** is discussed in Sec. 2.4.

Remark 2.5 (Role of Weingarten calculus)**.** The Weingarten formula of [8] enters the proof at the level of the resolvent expansion (6), not the exponential. The explicit Weingarten functions $\mathrm{Wg}(\sigma\tau^{-1}, L)$ for $\sigma, \tau \in S_n$ govern the $O(L^{-|\sigma\tau^{-1}|})$ corrections at each order of the moment expansion; resummation over non-crossing partitions recovers the subordination equation (7). For the orthogonal group $O(L)$, the Weingarten functions are replaced by zonal spherical functions [8], yielding a ramp coefficient twice that of GUE; for $Sp(L)$ the coefficient is halved, reproducing the full Wigner-Dyson universality classification.

2.4 Plateau and Heisenberg Time

The linear ramp of Eq. (9) saturates at the Heisenberg time $t_H = 2\pi N\rho_c$, set by the mean level spacing $\Delta = 1/(N\rho_c)$ of the compressed spectrum. Spectral rigidity [10] constrains the number variance $\Sigma^2(E) = \mathrm{Var}(\#\text{levels in } [E_0, E_0 + E])$ to grow only logarithmically for GUE, enforcing the plateau:

$$K_{\text{plateau}} = \frac{t_H}{2\pi} = N\rho_c, \quad t > t_H. \tag{10}$$

Because the scrambled subspace has finite dimension $N \sim e^{S_{\text{th}}}$, the level spacing is $\Delta \sim e^{-S_{\text{th}}}$, setting $t_H \sim e^{S_{\text{th}}}$ as expected from the black hole information paradox [2, 35].

3 Bipartite Entanglement and Page's Theorem

3.1 Bipartite Decomposition

We partition the total degrees of freedom into a local system S of dimension d and environment E such that $\mathcal{H}_{\text{total}} = \mathcal{H}_S \otimes \mathcal{H}_E$, where $\dim \mathcal{H}_E = N_E$. A generic initial state of S in the measurement pointer basis $\{|s_i\rangle\}_{i=1}^n$ takes the form $|\psi_S\rangle = \sum_{i=1}^n c_i|s_i\rangle$, with $\sum_i |c_i|^2 = 1$. Under the joint unitary evolution generated by $H(\alpha)$, the full state $|\Psi_{SE}\rangle$ evolves into the scrambled sector, entangling S with E.

3.2 Maximal Entropy via Page's Theorem

Theorem 3.1 (Page, 1993 [11])**.** *Let $|\Psi\rangle$ be a uniformly random pure state on a bipartite Hilbert space $\mathcal{H}_S \otimes \mathcal{H}_E$ with $\dim \mathcal{H}_S = d \leq \dim \mathcal{H}_E = N_E$. Then the average von Neumann entropy of*

the reduced density matrix $\rho_S = \mathrm{Tr}_E|\Psi\rangle\langle\Psi|$ *satisfies*

$$\langle S(\rho_S)\rangle = \ln d - \frac{d}{2N_E} + O\left(\frac{d^2}{N_E^2}\right). \tag{11}$$

In our setting, the kinematic constraint forces $|\Psi_{SE}\rangle$ to lie in a subspace with $N_E \sim e^{S_{\mathrm{th}}}$. Since S_{th} is the thermal entropy at the temperature scale set by H_{post}, we have $N_E \gg d$ for any macroscopic system, and the correction term $d/(2N_E)$ is exponentially suppressed:

$$\langle S(\rho_S)\rangle = \ln d - \frac{d}{2N_E} \xrightarrow{N_E\to\infty} \ln d \equiv S_{\max}. \tag{12}$$

Furthermore, Page's result is sharp: the standard deviation of $S(\rho_S)$ about its mean is $\sigma \sim d/(N_E \ln 2)$, which also vanishes exponentially. Therefore, the entropy is concentrated around its maximum with exponentially small fluctuations.

Corollary 3.2 (Kinematically Forced Maximally Mixed State)**.** *Under the kinematic constraint, the reduced density matrix* ρ_S *is exponentially close to the maximally mixed state:*

$$\left\|\rho_S - \frac{I_d}{d}\right\|_1 \leq \sqrt{\frac{d}{N_E}} \xrightarrow{N_E\to\infty} 0, \tag{13}$$

where $\|\cdot\|_1$ *denotes the trace norm. This bound follows from the purity formula implied by Page's theorem,* $\mathrm{Tr}(\rho_S^2) \approx 1/d + 1/N_E$*, combined with the Cauchy-Schwarz inequality for the trace norm:* $\|A\|_1 \leq \sqrt{d}\|A\|_2$*, giving* $\|\rho_S - I_d/d\|_1 \leq \sqrt{d}\|\rho_S - I_d/d\|_2 = \sqrt{\mathrm{Tr}(\rho_S^2) - 1/d} \approx \sqrt{1/N_E}\cdot\sqrt{d}^{1/2} \sim \sqrt{d/N_E}$.

The physical content of Corollary 3.2 is that decoherence into the scrambled environment is not merely efficient but kinematically guaranteed to be maximal. This provides the key precondition for the envariance argument that follows.

4 Envariance and the Derivation of Equal Probabilities

4.1 Axioms of Envariance

With maximal entanglement established by Corollary 3.2, we apply Zurek's environment-assisted invariance. The argument rests on three axioms that are operationally well-defined [14, 15]:

Definition 4.1 (Envariance Axioms)**.** Let $|\Psi_{SE}\rangle$ be a pure bipartite state. An operation u_S acting on $\mathcal{H}_S$ is called *envariant* if there exists a counter-operation $\tilde{u}_E$ on $\mathcal{H}_E$ such that

$$(u_S \otimes \tilde{u}_E)|\Psi_{SE}\rangle = |\Psi_{SE}\rangle. \tag{14}$$

The three axioms are:

(A1) **State-dependence of outcomes**: The probability of a measurement outcome depends only on the quantum state $|\Psi_{SE}\rangle$ and on the observable, not on the history by which the state was prepared.

(A2) **Causality / No-Signaling**: An envariant operation u_S acting on $\mathcal{H}_S$ cannot change any local observable of S. If u_S is envariant, then for any observable O_S on $\mathcal{H}_S$,

$$\mathrm{Tr}[O_S \rho_S] = \mathrm{Tr}[O_S (u_S \rho_S u_S^\dagger)]. \tag{15}$$

(A3) **Additivity**: For disjoint, mutually exclusive fine-grained events, probabilities are additive.

Axiom (A2) is the crucial physical input. It states that local facts about S cannot depend on operations performed on the causally disconnected E. This is precisely the no-signaling theorem in relativistic quantum field theory [19]: operations on E cannot propagate information to S outside the light cone. The envariance condition ensures that any transformation of S can in principle be undone by remote action on E; causality then forces the local state to be invariant under such transformations.

4.2 Proof of Equal Probabilities for Maximally Entangled States

Theorem 4.2 (Equal Probabilities from Envariance)**.** *Let $|\Psi_{SE}\rangle$ be a maximally entangled state of the form*

$$|\Psi_{SE}\rangle = \frac{1}{\sqrt{d}} \sum_{k=1}^{d} |s_k\rangle \otimes |e_k\rangle, \tag{16}$$

with $\{|s_k\rangle\}$ and $\{|e_k\rangle\}$ orthonormal bases of $\mathcal{H}_S$ and a d-dimensional subspace of $\mathcal{H}_E$ respectively. Then, under Axioms (A1)–(A3), the probability p_k of measuring the state $|s_k\rangle$ is $p_k = 1/d$ for all k.

Proof. Consider a swap unitary $U_S^{(jk)}$ that exchanges $|s_j\rangle \leftrightarrow |s_k\rangle$ and acts as the identity on all other basis vectors. Direct computation shows:

$$\left(U_S^{(jk)} \otimes U_E^{(jk)}\right) |\Psi_{SE}\rangle = \frac{1}{\sqrt{d}} \sum_{i=1}^{d} |s_{\sigma_{jk}(i)}\rangle \otimes |e_{\sigma_{jk}(i)}\rangle = |\Psi_{SE}\rangle, \tag{17}$$

where σ_{jk} is the transposition $(j\ k)$. Therefore $U_S^{(jk)}$ is envariant with counter-operation $\tilde{u}_E = U_E^{(jk)}$.

By Axiom (A2), the envariance of $U_S^{(jk)}$ implies that the action of $U_S^{(jk)}$ on $\mathcal{H}_S$ cannot alter any local property of S. In particular, since $U_S^{(jk)}$ maps $|s_j\rangle$ to $|s_k\rangle$, if the probabilities $p_j \neq p_k$, then the local distribution of outcomes would change under $U_S^{(jk)}$, which would be detectable by local measurements on S alone. But the envariant counter-operation $U_E^{(jk)}$ could then undo this change from E - implying that remote operations on E can set the outcome statistics in S, violating Axiom (A2). Therefore we must have $p_j = p_k$ for all j, k.

Since the probabilities are all equal and must sum to unity, $\sum_{k=1}^{d} p_k = 1$, we conclude $p_k = 1/d$ for all $k \in \{1, \ldots, d\}$. □ □

Remark 4.3 (Necessity of Maximal Entanglement)**.** The proof of Theorem 4.2 relies on the state (16) having equal Schmidt coefficients. For a non-maximally entangled state $|\Phi\rangle = \sum_k \lambda_k |s_k\rangle|e_k\rangle$ with $|\lambda_j| \neq |\lambda_k|$, the swap $U_S^{(jk)}$ is no longer envariant; its counter-operation would require complex amplitude manipulations in E, not a simple swap. The kinematic constraint of Sec. 3 is therefore essential: it ensures that the global state is exponentially close to a maximally entangled state, activating the envariance argument uniformly across all pointer states.

5 Coarse-Graining and the General Born Rule

5.1 Rational Approximation and Fine-Grained Branches

Theorem 4.2 gives equal probabilities only for maximally entangled states with equal Schmidt coefficients. To derive the general Born rule for arbitrary amplitudes c_i, we use the coarse-graining argument of [15], adapted to our kinematic setting.

Let $|\psi_S\rangle = \sum_{i=1}^{n} c_i |s_i\rangle$ with $|c_i|^2 = m_i/M$, where m_i are positive integers and $M = \sum_i m_i$. (The case of irrational $|c_i|^2$ is handled by a density argument in the Hilbert space topology,

as discussed in Sec. 5.3.) The key kinematic input is that the scrambled environment E has dimension $N_E \sim e^{S_{\rm th}} \gg M$, so the fine-grained partition is realizable.

Because the effective environmental Hamiltonian h_α is constructed via the uniform Haar-random isometry V_α, the N_E-dimensional Hilbert space of the environment is partitioned uniformly. The dynamical content of $H(\alpha)$ then forces the entanglement structure:

$$|\Psi_{SE}\rangle = \sum_{i=1}^{n} \sum_{j=1}^{m_i} \frac{c_i}{\sqrt{m_i}} |s_i\rangle \otimes |e_{i,j}\rangle, \tag{18}$$

where $\{|e_{i,j}\rangle\}$ is an orthonormal collection of M environmental states, with the m_i states $\{|e_{i,j}\rangle_{j=1}^{m_i}\}$ correlated with the pointer state $|s_i\rangle$.

The kinematic constraint forces $|c_i|/\sqrt{m_i} = 1/\sqrt{M}$ for all i, which is precisely the condition $|c_i|^2 = m_i/M$. This can be verified by noting that the Haar-random environment distributes Schmidt weights uniformly across the M fine-grained branches. Eq. (18) therefore simplifies to:

$$|\Psi_{SE}\rangle = \frac{1}{\sqrt{M}} \sum_{\mu=1}^{M} e^{i\phi_\mu} |s_\mu\rangle \otimes |e_\mu\rangle, \tag{19}$$

where μ is a fine-grained index running over all M branches, $|s_\mu\rangle$ denotes the coarse-grained pointer state $|s_i\rangle$ to which the μ-th fine-grained branch belongs, and $e^{i\phi_\mu}$ are arbitrary phases that do not affect the probabilities (as envariance extends to arbitrary unitary operations, not just swaps [15]).

5.2 Application of Envariance to Fine-Grained State

The fine-grained state (19) is exactly of the maximally entangled form (16) with d replaced by M. By Theorem 4.2, causality requires that every individual fine-grained branch μ is assigned equal probability:

$$p_\mu = \frac{1}{M}, \quad \forall\, \mu \in \{1, \ldots, M\}. \tag{20}$$

To obtain the probability of the coarse-grained macroscopic outcome $|s_i\rangle$, we sum over the m_i fine-grained branches correlated with that outcome. Using Axiom (A3):

$$p_i = \sum_{j=1}^{m_i} p_{i,j} = \frac{m_i}{M} = |c_i|^2. \tag{21}$$

The Born rule is thereby derived from the combined application of the kinematic constraint, the maximally entangled structure it enforces, and relativistic causality (no-signaling).

5.3 Extension to Irrational Amplitudes

The argument above assumed $|c_i|^2 \in \mathbb{Q}$. For the general case, we argue as follows. The set of states with rational squared amplitudes is dense in the unit sphere $\mathbb{S}^{2n-1} \subset \mathbb{C}^n$ in the topology induced by the ℓ^2 norm. For any state $|\psi_S\rangle$ with irrational $|c_i|^2$ and any $\epsilon > 0$, there exists a state $|\psi_S^\epsilon\rangle$ with rational squared amplitudes such that $\||\psi_S\rangle - |\psi_S^\epsilon\rangle\|_2 < \epsilon$. By Axiom (A1), the probability functional $p : \mathbb{S}^{2n-1} \to [0, 1]$ depends continuously on the state (a continuity assumption that is standard in quantum mechanics and follows from the requirement that small state perturbations lead to small measurement probability perturbations). Therefore, $p_i(|\psi_S\rangle) = \lim_{\epsilon\to 0} p_i(|\psi_S^\epsilon\rangle) = \lim_{\epsilon\to 0} |c_i^\epsilon|^2 = |c_i|^2$.

6 Objections and Replies

6.1 The Circularity Objection

Objection (Barnum et al. [16]): Envariance-based derivations of the Born rule are circular because they implicitly assume the Born rule in order to justify why the envariance transformation (14) leaves the probabilities invariant. Specifically, one must already know that two branches with equal Schmidt coefficients have equal probabilities in order to conclude that the swap operation is envariant in the probabilistically relevant sense.

Reply: This objection has genuine force when applied to envariance in isolation, but it does not apply in the present kinematic framework. Our argument for equal probabilities in Theorem 4.2 does not invoke the Born rule at any step. The envariance of the swap $U_S^{(jk)}$ is established by direct algebraic computation (17), which is a statement about the structure of the state vector, not about probabilities. The physical content that converts this algebraic fact into a probabilistic constraint is Axiom (A2) - the no-signaling theorem - which is an independent physical principle rooted in relativistic causality and the structure of quantum field theory. No probabilistic assumptions enter this step. The conclusion $p_j = p_k$ then follows purely from the impossibility of superluminal signaling.

Furthermore, the coarse-graining argument of Sec. 5 does not assume any particular probability rule to establish that the fine-grained state (19) has M equal Schmidt coefficients. This

follows from the specific kinematic constraint (3) and the arithmetic condition $|c_i|^2 = m_i/M$, which are established independently of any probabilistic considerations.

6.2 The Preferred Basis Problem

Objection: The derivation of the Born rule in Sec. 5 assumes a preferred pointer basis $\{|s_i\rangle\}$ for the system S. But in the bare Hilbert space formalism, no such basis is singled out a priori. Why should measurements occur in the pointer basis rather than in some other basis?

Reply: The kinematic framework provides a dynamical resolution of the preferred basis problem via decoherence [20, 21]. The pointer basis is not injected by hand but is selected by the structure of the interaction between S and E. Specifically, the pointer basis $\{|s_i\rangle\}$ is the basis that commutes with the interaction Hamiltonian H_{int} that entangles S with the environmental degrees of freedom governed by $H(\alpha)$:

$$[H_{\text{int}}, |s_i\rangle\langle s_i|] = 0, \quad \forall\, i. \tag{22}$$

States in this basis are the "pointer states" that decohere fastest into the scrambled environment; they are the stable attractors of the environmental monitoring process [22]. This basis is therefore determined by the dynamics of $H(\alpha)$, not by external stipulation. In macroscopic systems with many environmental degrees of freedom, the pointer basis is typically a classically distinguishable basis (position, momentum, particle number, etc.), consistent with empirical observation.

6.3 The Haar Measure Assumption

Objection: The entire kinematic framework rests on the ansatz that V_α is drawn from the Haar measure on G. Why should a physical system select a Haar-distributed subspace? Is this not simply relocating the mystery of quantum probability to the mystery of why the Haar measure is the correct measure on the space of scrambled states?

Reply: We acknowledge this as the principal open question of the framework. However, there are several strong arguments in favor of the Haar measure as the correct kinematic ensemble:

First, the Haar measure is the *unique* unitarily invariant measure on G. Any other measure would break the rotational symmetry of the Hilbert space, introducing a preferred direction

or preferred subspace that would itself require explanation. The Haar measure is the minimal assumption consistent with the symmetry of the underlying Hilbert space.

Second, the Haar measure can be justified from a maximum entropy argument [25]: given no information about the preferred subspace other than its dimension N, the maximum entropy distribution over N-dimensional subspaces is exactly the Haar measure on the Grassmannian $\mathrm{Gr}(N, L)$, to which the Haar measure on $U(L)$ descends.

Third, recent developments in quantum gravity and holography provide microscopic evidence for Haar-random scrambling. In the Hayden-Preskill thought experiment [6], the late-time dynamics of an old black hole are modeled by a Haar-random unitary on the full Hilbert space of Hawking radiation. This Haar-randomness is not imposed by hand but follows from the requirement of information-theoretic unitarity and the monogamy of entanglement. Moreover, the JT gravity path integral [4] computes the SFF via a sum over topologies that precisely reproduces Sec. 2, providing a non-perturbative gravitational justification for the kinematic ansatz.

Fourth, for generic many-body systems, the eigenstate thermalization hypothesis (ETH) [23, 24] implies that the matrix elements of local observables in the energy eigenbasis are Gaussian-distributed with variance $e^{-S_{\mathrm{th}}/2}$, consistent with Haar-random eigenvectors. This provides a many-body field theory argument for the Haar measure assumption in the thermodynamic limit.

6.4 Comparison with Alternative Derivations of the Born Rule

Several alternative approaches to deriving the Born rule have been proposed in the literature. It is instructive to position the present framework relative to these alternatives.

Decision-theoretic derivations (Deutsch-Wallace [26, 27]): These approaches, developed in the context of the Everett (many-worlds) interpretation, derive the Born rule from rational agency: an agent who assigns unequal probabilities to equal-coefficient branches would be behaving irrationally according to the axioms of decision theory. While elegant, these approaches have been criticized for circularity [28, 29]: the rationality axioms used implicitly assume a probabilistic framework that presupposes the Born rule. In contrast, the present framework derives probabilities from a non-probabilistic physical principle (no-signaling / causality), avoiding this circularity.

Typicality arguments (Everett, Deutsch): Typicality arguments attempt to show that, in the limit of many identical measurements, branches with the Born rule measure are "typical"

in some sense. These arguments face the challenge of defining typicality without invoking probability [30]. Our approach sidesteps this by deriving the Born rule from a single measurement setting.

Operational approaches (QBism, Rovelli): Relational quantum mechanics [31] and QBism [32] treat probabilities as irreducibly subjective or relational, not derivable from the formalism. These approaches effectively accept the Born rule as a primitive. Our framework is more ambitious in attempting a dynamical derivation.

Pilot wave theories (Bohmian mechanics [33]): In Bohmian mechanics, particles follow deterministic trajectories guided by the wave function, and the Born rule is derived from the quantum equilibrium hypothesis (the assumption that the particle distribution is $|\psi|^2$ in equilibrium). Our framework is complementary: we also derive the Born rule from an equilibrium distribution, but our equilibrium distribution is over environmental microstates rather than particle positions. Both approaches relocate the probabilistic burden; our relocation is to a classical geometric object (the Haar measure on a matrix group).

6.5 Unitarity and the Block-Diagonal Structure

Objection: The block-diagonal structure (2) of $H(\alpha)$ appears to prevent coherent superpositions between the early-time sector $(I - \Pi_\alpha)$ and the late-time sector Π_α. Does this not violate unitarity by effectively collapsing the state at the Thouless time?

Reply: The block-diagonal form of $H(\alpha)$ does not violate unitarity. It is a kinematic restriction on which transitions are *dynamically accessible* at late times, not a physical collapse of the wave function. Formally, $H(\alpha)$ is a self-adjoint operator on $\mathcal{H}$, and $e^{-iH(\alpha)t}$ is a unitary transformation for all t. The block-diagonal structure implies that states initialized in the early-time sector $(I - \Pi_\alpha)\mathcal{H}$ evolve into the late-time sector $\Pi_\alpha\mathcal{H}$ via the overlapping action of H_{base}, but do not coherently return. This is physically analogous to the one-way flow of information in thermodynamic systems: unitary evolution can drive a state from a low-entropy sector to a high-entropy sector without ever returning, consistent with the arrow of time. The scrambling dynamics encoded in Π_α are the many-body manifestation of this thermodynamic arrow.

7 Discussion: Shifting the Probabilistic Burden

7.1 From Quantum Axiom to Classical Geometry

A common and valid objection to kinematic derivations of quantum probability is that probability must enter the theory at some level. In the framework presented here, we do not claim to banish probability from physics; rather, we systematically relocate its origin along three conceptual steps.

In standard quantum mechanics, probability enters as a first-principles axiom: the Born rule is postulated as an irreducible, a priori element of the formalism with no dynamical basis. In our framework:

1. **Step 1 (Quantum → Kinematic):** We replace the Born rule (a quantum axiom) with the Haar measure (a classical, geometric measure on a matrix group). The Haar measure is the unique invariant volume form on G, entirely analogous to the uniform measure on a sphere or the Liouville measure on phase space in classical statistical mechanics. It is not mysterious in the same way as quantum probability.

2. **Step 2 (Kinematic → Thermodynamic):** The Haar measure can be further grounded in the maximum entropy principle: Haar-randomness is the result of maximizing the Boltzmann entropy over all unitarily invariant distributions on the space of N-dimensional subspaces of $\mathcal{H}$.

3. **Step 3 (Thermodynamic → Gravitational):** As discussed in Sec. 6.3, microscopic justification for the Haar measure arises from non-perturbative quantum gravity (JT gravity, black hole thermalization), the ETH, or Hayden-Preskill scrambling. In all cases, the Haar measure emerges as the natural late-time distribution over quantum states.

7.2 Connections to Black Hole Information

The kinematic framework has a natural interpretation in the context of the black hole information paradox [7]. The late-time scrambled sector $\Pi_\alpha \mathcal{H}$ can be identified with the Hilbert space of Hawking radiation after the Page time [12]. The Haar-random isometry Φ_α represents the unitary black hole evaporation map, which scrambles the infalling information into Hawking radiation. The Born probabilities for a local observer outside the black hole then correspond to

the probabilities assigned to fine-grained radiation microstates by the envariance argument of Sec. 5.

In this picture, the information paradox is resolved kinematically: the apparent mixed state of Hawking radiation is not a fundamental failure of unitarity but a coarse-grained description obtained by summing over exponentially many fine-grained branches, each contributing with equal weight $1/M$ by Eq. (20). The entanglement entropy of the radiation follows the Page curve [12] by construction, consistent with recent calculations in doubly holographic models [34].

7.3 Limitations and Open Questions

We emphasize that the success of this derivation hinges on the validity of the kinematic ansatz: the assumption that the late-time scrambled subspace is Haar-random. While this assumption recovers the SFF dip-ramp-plateau and the Born rule, its ultimate justification depends on:

1. A non-perturbative proof of Haar-random scrambling from microscopic many-body dynamics, presumably within the ETH framework.

2. A microscopic quantum gravitational derivation of the kinematic constraint, e.g., from the matrix model dual of the JT gravity path integral.

3. An extension to systems with conserved charges (charge, energy, angular momentum), where the Haar measure on $U(N)$ should be replaced by a suitable measure on the constrained unitary group.

4. A precise characterization of the Thouless time t_{Th}, before which the kinematic constraint is not yet in force and the Born rule derivation does not apply. This requires a microscopic treatment of the early-time $(I - \Pi_\alpha)$ sector.

8 Conclusion

We have presented a kinematic framework in which the two fundamental mysteries of random matrix universality in quantum chaos and the Born rule of quantum measurement admit a unified resolution. By postulating a Haar-random kinematic constraint on the late-time scrambled sector, we have reproduced the SFF dip-ramp-plateau structure.

This same kinematic structure provides for deriving the Born rule. Page's theorem guarantees that the local system is driven exponentially close to a maximally mixed state, activating the envariance argument. A self-contained proof shows that relativistic causality - the no-signaling theorem - forces equal probabilities on all fine-grained branches of the maximally entangled state. The coarse-graining argument then yields $p_k = |c_k|^2$ for arbitrary initial amplitudes, including the irrational case by density.

We have systematically addressed the major objections to this program. The circularity objection does not apply because the key step (Theorem 4.2) invokes no probabilistic assumption; it relies solely on the algebraic structure of the entangled state and the physical content of no-signaling. The preferred basis problem is resolved by decoherence theory. The Haar measure assumption, while the principal open question, is supported by the uniqueness of the Haar measure under rotational invariance, the maximum entropy principle, ETH, and non-perturbative quantum gravity.

In this view, quantum probability is an emergent, coarse-grained kinematic consequence of entanglement, causality, and the classical geometry of the Haar measure over the space of scrambled states. The probabilistic nature of quantum mechanics is not a fundamental feature of reality but a consequence of our description of a small local system embedded in an exponentially large, Haar-randomly scrambled environment.

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